Build With Claude — Day 2

Building a Product with Claude Code: Skills, Subagents & Plugings

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This lab packages two askdb skills as **one** Claude Code plugin, publishes it from a GitHub repository, and adds it to an askdb Claude Code project.

Skill	Command after install
`create-db-configs`	`/askdb:create-db-configs`
`create-eval-dataset`	`/askdb:create-eval-dataset`

The plugin is named `askdb`. The marketplace (the GitHub repository) is named `askdb-skills`. You install the plugin with `askdb@askdb-skills`.

The skill files already exist. You will copy them into the folders of the plugin repository.

Two sets of commands. Every step that differs between systems shows two blocks:

- **macOS / Linux (Terminal):** run in the macOS Terminal or a Linux terminal.
- **Windows (PowerShell):** run in Windows PowerShell or PowerShell 7.

Where a command is the same on every system (for example `claude`, `git`, `gh`, `uv`), it is shown once.

Folders. The commands in this guide do not change folders. Each step tells you which folder to be in (for example, "Go to the `askdb-skills` folder"). Open a terminal in that folder before you run the commands in the step.

You will:

1. Check your prerequisites and create the repository folder.
2. Create the folders and copy the files into the plugin repository.
3. Validate the plugin and the marketplace.
4. Test the plugin on your machine.
5. Publish the repository on GitHub.
6. Add the plugin to an askdb Claude Code project.
7. Use the skills, release an update, and undo the setup if needed.

Plugin structure reference: https://code.claude.com/docs/en/plugins/create. This lab was tested on Claude Code **2.1.312** on macOS.

> **Windows note.** The Windows commands in this guide follow PowerShell syntax. They were written for Windows PowerShell 5.1 (included with Windows) and PowerShell 7, but they have not been run on Windows. If a Windows command fails, read the error and see the Troubleshooting table.

Part 0. Prerequisites

Run each check. Do not skip one.

0.1 Open a terminal

- **macOS:** Applications > Utilities > Terminal.
- **Linux:** any terminal.
- **Windows:** click Start, type `PowerShell`, and open **Windows PowerShell**. You can also use **PowerShell 7** if it is installed.

0.2 Check Claude Code

Same on every system:

```
claude --version
```

Expected: a version such as `2.1.312` or higher. If the command is not found, install Claude Code and sign in with `claude`:

- macOS and Linux: https://code.claude.com/docs/en/quickstart
- Windows: follow the Windows instructions on the same page.

0.3 Check git

Same on every system:

```
git --version
```

Expected: `git version 2.x.x`. If it is missing:

- macOS: install git from https://git-scm.com/downloads (or run `xcode-select --install`).
- Windows: install git from https://git-scm.com/downloads, or run `winget install --id Git.Git -e` in PowerShell. Close and reopen PowerShell afterwards.

0.4 Check the GitHub CLI and sign in

Same on every system:

```
gh --version
gh auth status
```

Expected: a version number, and `Logged in to github.com`. If you are not logged in:

```
gh auth login
```

Choose **GitHub.com**, then **HTTPS**, and follow the browser prompts. Run `gh auth status` again.

If `gh` is missing:

- macOS: `brew install gh`, or the installer at https://cli.github.com.

- Windows: `winget install --id GitHub.cli -e`, or the installer at https://cli.github.com. Close and reopen PowerShell afterwards.

0.5 Check uv

The skills run their scripts with `uv`. Same on every system:

```
uv --version
```

Expected: `uv 0.x.x`. If it is missing:

- macOS and Linux: follow https://docs.astral.sh/uv/getting-started/installation/.
- Windows: `winget install --id astral-sh.uv -e`, or follow the same page. Close and reopen PowerShell afterwards.

0.6 Create the repository folder

Create an empty folder named `askdb-skills` in a place you choose, such as your Documents folder.

Then open a terminal inside that folder:

- **macOS:** in Finder, right-click the `askdb-skills` folder and choose **New Terminal at Folder**. If that option is not there, open Terminal and drag the folder onto its window.
- **Linux:** right-click the `askdb-skills` folder and choose **Open in Terminal**, or the equivalent option in your file manager.
- **Windows:** open the `askdb-skills` folder in File Explorer, click the address bar, type `powershell`, and press Enter.

Check that the terminal is in the right folder:

macOS / Linux (Terminal):

```
pwd
```

Windows (PowerShell):

```
Get-Location
```

Expected: the last part of the path is `askdb-skills`. Keep this terminal open for Parts 2 and 3.

0.7 Know where the source files are

You need the skill files and the three metadata files listed in Part 2. Your instructor or the lab handout tells you where they are. Note that folder before you start Part 2. In this guide, `<SOURCE>` means that folder.

Part 1. Understand the target layout

The repository is the plugin. Its root holds:

- `.claude-plugin/plugin.json`: the plugin manifest (name, description, version).
- `.claude-plugin/marketplace.json`: the catalog that lists the plugin, so people can install it by name.

- `skills/<skill-name>/SKILL.md` : one folder per skill. Both skills go under this one `skills/` folder.
- `skills/<skill-name>/scripts/...` and `skills/<skill-name>/references/...` : the files each skill uses.
- `README.md` : what the plugin does.

The finished repository looks like this:

```
askdb-skills/                                  <- GitHub repository = the plugin
├── .claude-plugin/
|   ├── plugin.json                            <- plugin name "askdb"
|   └── marketplace.json                       <- marketplace name "askdb-skills"
├── README.md
├── LAB_GUIDE.md                               <- optional: this guide
└── skills/
    ├── create-db-configs/
    |   ├── SKILL.md
    |   └── scripts/
    |       └── askdb_config.py
    └── create-eval-dataset/
        ├── SKILL.md
        ├── references/
        |   └── sample_eval.yaml
        └── scripts/
            └── check_eval_set.py
```

Windows paths use backslashes (`C:\Users\you\askdb-skills`). PowerShell also accepts forward slashes, but the commands below use backslashes on Windows to match Windows paths.

Part 2. Create the folders and copy the files

Stay in the `askdb-skills` folder for all steps in this part.

Step 2.1 Create the folders

Create the plugin folder and the skill folders. The commands below create the parent folders too.

macOS / Linux (Terminal):

```
mkdir -p .claude-plugin
mkdir -p skills/create-db-configs/scripts
mkdir -p skills/create-eval-dataset/references
mkdir -p skills/create-eval-dataset/scripts
```

Windows (PowerShell):

```
md .claude-plugin
md skills\create-db-configs\scripts
md skills\create-eval-dataset\references
md skills\create-eval-dataset\scripts
```

Expected: no error. If Windows prints `already exists` , the folder is already there. That is fine.

Step 2.2 Copy the files

Copy each file from `<SOURCE>` to the folder in the table. Keep the file name the same. Two files are named `SKILL.md`, so check the skill name in the first column.

#	Source file	Copy to (inside `askdb-skills`)
1	`SKILL.md` for create-db-configs	`skills/create-db-configs/`
2	`askdb_config.py`	`skills/create-db-configs/scripts/`
3	`SKILL.md` for create-eval-dataset	`skills/create-eval-dataset/`
4	`sample_eval.yaml`	`skills/create-eval-dataset/references/`
5	`check_eval_set.py`	`skills/create-eval-dataset/scripts/`
6	`plugin.json`	`.claude-plugin/`
7	`marketplace.json`	`.claude-plugin/`
8	`README.md`	the `askdb-skills` folder itself

You can drag and drop the files in Finder or File Explorer, or use one command per file:

macOS / Linux (Terminal):

```
cp "<SOURCE>/SKILL.md" skills/create-db-configs/
```

Windows (PowerShell):

```
Copy-Item "<SOURCE>\SKILL.md" skills\create-db-configs\
```

Replace the file name and the folder for each row of the table. Replace `<SOURCE>` with the folder from Step 0.7.

Content of rows 6 to 8

Rows 6 to 8 are the three metadata files. Copy them from `<SOURCE>` as in the table. If these three files were not given to you, create them with the content below. Keep the text exactly as shown.

Row 6: `.claude-plugin/plugin.json`

```
{
  "name": "askdb",
  "description": "Skills for askdb: create database YAML configs for SQLite
databases, and create evaluation datasets with confirmed expected SQL.",
  "version": "1.0.0",
  "author": {
    "name": "Manaranjan Pradhan"
  }
}
```

Row 7: `.claude-plugin/marketplace.json`

```
{
  "name": "askdb-skills",
  "owner": {
    "name": "Manaranjan Pradhan"
  },
  "metadata": {
    "description": "Claude Code plugin for askdb: database config and evaluation
dataset skills."
  },
  "plugins": [
    {
      "name": "askdb",
      "source": "./",
      "description": "Create askdb database configs and evaluation datasets."
    }
  ]
}
```

Row 8: `README.md`

```
# askdb plugin

One plugin with two skills for askdb.

| Skill | Command | What it does |
| :- | :- | :- |
| create-db-configs | `/askdb:create-db-configs <sqlite-file>` | Creates the askdb
database YAML for a SQLite file |
| create-eval-dataset | `/askdb:create-eval-dataset <sqlite-file> <config-yaml>` |
Creates an evaluation set with confirmed expected SQL |

## Requirements

- The Claude Code project must be the askdb repository, or a checkout of it,
because both skills run `uv run` against the `askdb` package.
- SQLite databases only.
```

Change `"Manaranjan Pradhan"` to your name in the first two files before publishing.

Step 2.3 Check the files

Still in the `askdb-skills` folder:

macOS / Linux (Terminal):

```
find . -type f -not -path './.git/*' | sort
```

Windows (PowerShell):

```
Get-ChildItem -Recurse -File | Select-Object -ExpandProperty FullName | Sort-Object
```

Expected: all eight files from the table are listed, in the folders shown in the table. The `.git` folder is not created yet, so it does not appear.

Check that no file still uses the old skill path:

macOS / Linux (Terminal):

```
grep -rn '\.claude/skills' . || echo "OK - no old paths left"
```

Windows (PowerShell):

```
$hits = Get-ChildItem -Recurse -File | Select-String -Pattern '\.claude[\\/]skills'
if ($hits) { $hits } else { "OK - no old paths left" }
```

Expected: `OK — no old paths left`. If a file is listed, ask for a corrected copy of that file. Do not edit it by hand.

Part 3. Validate the plugin and the marketplace

Stay in the `askdb-skills` folder. Run the validator with `--strict`, which also fails on warnings. The command is the same on every system except the way the path is written.

Step 3.1 Validate the plugin

macOS / Linux:

```
claude plugin validate --strict .
```

Windows:

```
claude plugin validate --strict .
```

Expected: `✔ Validation passed`. You may also see advice about a missing `LICENSE` or `icon`. Those matter only if you submit to Anthropic's public directory. They do not fail the check.

Step 3.2 Check the exit code

macOS / Linux:

```
claude plugin validate --strict . >/dev/null 2>&1; echo "exit: $?"
```

Windows:

```
claude plugin validate --strict . | Out-Null; "exit: $LASTEXITCODE"
```

Expected: `exit: 0`.

If a check fails, read the message above the result line. It names the file and the field to fix.

Part 4. Test the plugin before you publish

Step 4.1 Load the plugin in a session

`--plugin-dir` loads the plugin for one session without installing it. You need the full path of the `askdb-skills` folder.

- **macOS:** in Finder, drag the `askdb-skills` folder onto the Terminal window. The full path is pasted for you.
- **Windows:** in File Explorer, open the `askdb-skills` folder, click the address bar, and copy the path.

Go to the askdb project folder. Then start Claude Code and load the plugin. Replace `<path to askdb-skills>` with the full path you copied:

macOS / Linux:

```
claude --plugin-dir "<path to askdb-skills>"
```

Windows:

```
claude --plugin-dir "<path to askdb-skills>"
```

In the session, type:

```
/plugin
```

Open the **Installed** tab. Expected: `askdb` is listed, with 2 skills, and the **Errors** tab is empty.

Type `/` and look for:

- `/askdb:create-db-configs`
- `/askdb:create-eval-dataset`

Exit with `/exit`.

> If you change a file while the session is open, type `/reload-plugins` in the session.

Part 5. Publish the repository on GitHub

Stay in the `askdb-skills` folder for all steps in this part.

Step 5.1 Make the folder a git repository

macOS / Linux and Windows:

```
git init -b main
```

Expected: `Initialized empty Git repository in .../askdb-skills/.git/`.

Step 5.2 Commit the files

The attribution line is optional. Keep it or remove it. The same commands work on every system:

```
git add .
git commit -m "Add askdb plugin with two skills" -m "Co-Authored-By: Claude Haiku
5.5 <noreply@anthropic.com>"
```

If git asks for your name or email, set them once and commit again:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Step 5.3 Create the GitHub repository and push

Choose a visibility:

- `--private`: only people you invite can see it. Recommended for a team.
- `--public`: anyone can see it.

Replace `<OWNER>` with your GitHub user name or organization:

```
gh repo create <OWNER>/askdb-skills --private --source . --remote origin --push
```

Expected: the command prints `https://github.com/<OWNER>/askdb-skills`. Open it in a browser and check that the files from Step 2.3 are there.

If the name is taken, choose another repository name. The install commands still use the marketplace name `askdb-skills`, because that name comes from `marketplace.json`.

Step 5.4 Check the remote

```
git remote -v
git status
```

Expected: `origin` points to your GitHub repository, and `git status` shows `nothing to commit, working tree clean`.

Part 6. Add the plugin to an askdb Claude Code project

Go to the askdb project folder for all steps in this part. Project scope writes the setting into `.claude/settings.json` in that project. You commit that file, so teammates get the same setup.

Step 6.1 Add the marketplace to the project

```
claude plugin marketplace add <OWNER>/askdb-skills --scope project
```

Expected: `✔ Successfully added marketplace: askdb-skills (declared in project settings)`.

For a private repository, your GitHub login must work on this machine. Check it with `gh auth status`.

Step 6.2 Install the plugin into the project

```
claude plugin install askdb@askdb-skills --scope project
```

Expected: `✔ Successfully installed plugin: askdb@askdb-skills (scope: project)`.

Step 6.3 Confirm the installation

macOS / Linux:

```
claude plugin details askdb
claude plugin list
cat .claude/settings.json
```

Windows:

```
claude plugin details askdb
claude plugin list
Get-Content .claude\settings.json
```

Expected: `details` lists the two skills, `create-db-configs` and `create-eval-dataset`. `list` shows `askdb@askdb-skills` with `Scope: project` and `Status: ✔ enabled`. The settings file has `"askdb@askdb-skills": true` under `enabledPlugins`, and an `askdb-skills` entry under `extraKnownMarketplaces`.

Step 6.4 Commit the project setting

Check the status first. Only `.claude/settings.json` should be staged.

```
git status
git add .claude/settings.json
git commit -m "Enable askdb plugin for this project" -m "Co-Authored-By: Claude
Haiku 5.5 <noreply@anthropic.com>"
git push
```

Step 6.5 Set up each teammate

The committed setting turns the plugin on for teammates, but it does not install it on their machines. Each teammate goes to the folder of their own clone of the askdb project, then runs these commands once:

```
claude plugin marketplace add <OWNER>/askdb-skills --scope project
claude plugin install askdb@askdb-skills --scope project
claude plugin list
```

Part 7. Use the skills

Go to the askdb project folder, then start Claude Code:

```
claude
```

Type `/` and check for `/askdb:create-db-configs` and `/askdb:create-eval-dataset`.

Run the first skill with a SQLite file:

```
/askdb:create-db-configs /path/to/your.sqlite
```

On Windows, use a path such as `C:\path\to\your.sqlite`.

Run the second skill with the file and the config YAML that the first skill wrote:

```
/askdb:create-eval-dataset /path/to/your.sqlite /path/to/your.yaml
```

Expected: the skill inspects the database and asks you questions. Answer them. Confirm each expected answer when it asks.

Part 8. Release an update

1. Go to the `askdb-skills` folder.
2. Edit files under the `skills` folder.
3. Increase `version` in `.claude-plugin/plugin.json`, for example from `1.0.0` to `1.0.1`. Claude Code sends an update only when the version changes.
4. Validate, commit, and push. The commands are the same on every system:

```
claude plugin validate --strict .
git add .
git commit -m "Release askdb 1.0.1" -m "Co-Authored-By: Claude Haiku 5.5
<noreply@anthropic.com>"
git push
```

5. Each user updates their copy. Go to the askdb project folder, then run:

```
claude plugin update askdb@askdb-skills
```

Then start a new session, or run `/reload-plugins` in an open session.

If `claude plugin update` says the plugin is already at the latest version, you did not increase `version`.

Part 9. Undo the setup

Go to the askdb project folder, then remove the plugin from the project:

```
claude plugin uninstall askdb@askdb-skills --scope project
```

Expected: `✔ Successfully uninstalled plugin: askdb (scope: project)`.

Remove the marketplace. The command is the same on every system:

```
claude plugin marketplace remove askdb-skills
```

Expected: `✔ Successfully removed marketplace: askdb-skills`.

If you had committed `.claude/settings.json`, commit the removal in the askdb project.

Appendix A. Field reference for the JSON files

JSON does not allow comments, so the explanation of each field is here. Keep the JSON files free of notes. The fields used in this lab are marked **Used**.

A.1 `.claude-plugin/plugin.json` (the plugin manifest)

Copied in Step 2.2 (row 6). The full content is shown in Step 2.2.

Field	Required	Used	What it means	Value in this lab
`name`	Yes	Yes	Plugin identifier. It becomes the prefix of every skill command (`/askdb:...`) and the name you type after `@` when installing. Use kebab-case, no spaces. Do not rename it after release.	`"askdb"`
`description`	No	Yes	One sentence shown to users in `/plugin` and in `claude plugin details`.	Skills for askdb: database config and evaluation dataset skills.
`version`	No	Yes	Release number. Claude Code sends an update only when this changes. Increase it on every release.	`"1.0.0"`
`author`	No	Yes	Object with the author's details.	See below
`author.name`	Yes, inside `author`	Yes	The author's name, shown to people who install the plugin.	`"Manaranjan Pradhan"`: change it to your name
`author.email`	No	No	Contact email for the author.	Not set
`author.url`	No	No	A web page for the author.	Not set
`displayName`	No	No	A label to show in place of `name`. Changing it does not affect installs.	Not set
`homepage`	No	No	A URL for the plugin's home page. It must be a valid URL.	Not set
`repository`	No	No	URL of the source repository.	Not set
`license`	No	No	SPDX license identifier, such as `MIT` or `Apache-2.0`. Required only for the public directory.	Not set (optional for a team repository; add it before a public release)

Rules that apply to every field:

- The file must be valid JSON. No trailing commas, and every string uses double quotes.
- Only `plugin.json` goes in `.claude-plugin/`. The skills go in `skills/`, at the plugin root.

A.2 `.claude-plugin/marketplace.json` (the catalog)

Copied in Step 2.2 (row 7). The full content is shown in Step 2.2.

Field	Required	Used	What it means	Value in this lab
`name`	Yes	Yes	Marketplace identifier. It is the part after `@` in every install command: `askdb@askdb-skills`. Do not change it after people have installed from it.	`"askdb-skills"`
`owner`	Yes	Yes	Object with the owner's details.	See below
`owner.name`	Yes, inside `owner`	Yes	The name of the person or team that owns the marketplace.	`"Manaranjan Pradhan"`: change it to your name
`metadata`	No	Yes	Object with extra details about the marketplace.	See below
`metadata.description`	No	Yes	One sentence about the marketplace. Without it, `claude plugin validate --strict` warns.	Claude Code plugin for askdb: database config and evaluation dataset skills.
`plugins`	Yes	Yes	List of plugin entries. Each entry describes one plugin in this repository.	One entry, below
`plugins[].name`	Yes	Yes	Must match `name` in that plugin's `plugin.json`.	`"askdb"`
`plugins[].source`	Yes	Yes	Where the plugin's files are. `"./"` means the repository root. A folder path such as `"./plugins/askdb"` is used when the plugin sits in a subfolder.	`"./"`
`plugins[].description`	No	Yes	One sentence shown in the **Discover** list before a user installs.	Create askdb database configs and evaluation datasets.
`renames`	No	No	Map from an old plugin name to a new one. Used only when you rename a published plugin, so existing installs migrate.	Not set

A.3 Steps and the files they touch

Step	Action (macOS/Linux / Windows)	Folder or file	Check
0.6	Create the folder and open a terminal in it	`askdb-skills` folder	`pwd` / `Get-Location` shows `askdb-skills`
0.7	Note the folder of the source files	Nothing	You know where `<SOURCE>` is
2.1	`mkdir -p` / `md`	Folders inside `askdb-skills`	No error
2.2	Copy (drag and drop, `cp` / `Copy-Item`)	Eight files, rows 1 to 8 of the table	Each file is in the folder shown
2.3	`find` / `Get-ChildItem`, `grep` / `Select-String`	Nothing	Eight files listed; no `.claude/skills` path
3.1 to 3.2	`claude plugin validate --strict .`	Nothing	Exit code 0 (`$LASTEXITCODE` on Windows)
4.1	`claude --plugin-dir` in the askdb project	Nothing	`askdb` listed under **Installed**
6.1	`claude plugin marketplace add ... --scope project`	`.claude/settings.json` in the askdb project	`extraKnownMarketplaces` entry exists
6.2	`claude plugin install askdb@askdb-skills --scope project`	`enabledPlugins` entry in `.claude/settings.json`	`claude plugin list` shows `enabled`

Troubleshooting

What you see	What to do
`✘ Validation failed` from `claude plugin validate`	Read the line above the result. It names the file and field. Fix it and run the check again.
`Plugin "askdb" not found in marketplace`	Check that `name` in `plugin.json` is `askdb`, and that `marketplace.json` lists it with `"source": "./"`.
`ModuleNotFoundError: No module named 'askdb'`	You are not in the askdb project folder. Go to the askdb project folder, run `uv sync` there, and run the command again.
`command not found: uv` (macOS/Linux) or `uv is not recognized` (Windows)	Install uv (Step 0.5). On Windows, close and reopen PowerShell after installing.
`cannot be loaded because running scripts is disabled` (Windows)	Run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, answer `Y`, then open a new PowerShell window and try again. This changes only your user account.
`md : Access is denied` (Windows)	Create the `askdb-skills` folder inside your user profile, such as `Documents`, and do not use `C:\Program Files`.
A file is missing or in the wrong folder after Step 2.2	Copy it again to the folder shown in the table. Check the file name, especially for the two `SKILL.md` files.
`.claude/skills` path found in Step 2.3	Ask for a corrected copy of that file. Do not edit it by hand.
`pwd` or `Get-Location` shows the wrong folder (Step 0.6)	Close the terminal, open it again inside the `askdb-skills` folder, and check again.
Skills do not appear after install	Run `/reload-plugins` in the session. Open `/plugin` and check the **Errors** tab. Run `claude plugin list` in your shell.
Adding a private marketplace fails	Run `gh auth status`. If needed, run `gh auth login`, then try again.
`already at the latest version` on update	Increase `version` in `plugin.json`, then commit and push (Part 8).

Checklist

- [ ] Prerequisites pass, the `askdb-skills` folder is created, and you know where the source files are (Part 0)
- [ ] Folders exist (Step 2.1)
- [ ] All eight files are copied to the folders in the table (Step 2.2)
- [ ] The check in Step 2.3 lists eight files and finds no old path
- [ ] `claude plugin validate --strict .` exits with 0 (Part 3)
- [ ] `/askdb:create-db-configs` and `/askdb:create-eval-dataset` appear in a `--plugin-dir` session (Step 4.1)
- [ ] The repository is on GitHub with the expected files (Part 5)
- [ ] `askdb@askdb-skills` is installed in the askdb project (Part 6)
- [ ] `.claude/settings.json` is committed in the askdb project (Step 6.4)